

Chapter 9: Hypothesis Testing

Sections 9.1–9.3

Critical Value Approach Only

MAT 1272 Statistics

Today's Big Idea

In Chapter 9, we use sample data to test a claim about a population parameter.

Examples:

- ▶ Does a soda can contain 12 ounces on average?
- ▶ Is the mean waiting time more than 10 minutes?
- ▶ Is the mean age at which children start walking different from 12.5 months?

We will focus on the **critical value approach**.

What We Are Covering

Section 9.1

Introduction to hypothesis testing.

Section 9.2

Tests about μ when σ is known.

Only critical value approach.

Section 9.3

Tests about μ when σ is unknown.

Only critical value approach.

Hypothesis Testing

A hypothesis test helps us decide whether sample evidence is strong enough to reject a claim about a population.

The claim is usually about a population parameter such as

$$\mu$$

the population mean.

Important: hypotheses are written using population parameters, not sample statistics.

Court Trial Analogy

Court Trial	Hypothesis Test
Person is not guilty	H_0
Person is guilty	H_a
Evidence	Sample data
Verdict	Statistical decision

We begin by assuming the null hypothesis is true.

Null Hypothesis

Null Hypothesis

The null hypothesis, denoted by H_0 , is the statement assumed to be true unless the sample evidence is strong enough to reject it.

The null hypothesis contains equality:

$$=, \leq, \geq$$

Alternative Hypothesis

Alternative Hypothesis

The alternative hypothesis, denoted by H_a , is the statement we support when the evidence against H_0 is strong.

The alternative hypothesis contains:

$<$, $>$, $\neq$

Example: Soda Cans

A company claims its cans contain 12 ounces of soda on average.

If we want to test whether the cans contain less than 12 ounces on average, then:

$$H_0 : \mu = 12$$

$$H_a : \mu < 12$$

This is a left-tailed test.

Important Language

At the end of a hypothesis test, we say either:

Reject H_0

or

Fail to reject H_0

Do not say:

Accept H_0

Failing to reject H_0 does not prove that H_0 is true.

Rejection and Nonrejection Regions

The critical value separates the rejection region from the nonrejection region.

- ▶ If the test statistic falls in the rejection region, reject H_0 .
- ▶ If the test statistic does not fall in the rejection region, fail to reject H_0 .

Type I Error

Type I Error

A Type I error occurs when we reject a true null hypothesis.

$$\alpha = P(\text{Reject } H_0 \mid H_0 \text{ is true})$$

The value of α is called the significance level.

Type II Error

Type II Error

A Type II error occurs when we fail to reject a false null hypothesis.

$$\beta = P(\text{Fail to reject } H_0 \mid H_0 \text{ is false})$$

This means there was a real difference, but the test did not detect it.

Four Possible Outcomes

	H_0 true	H_0 false
Fail to reject H_0	Correct decision	Type II error
Reject H_0	Type I error	Correct decision

The two errors are different, and students should not mix them up.

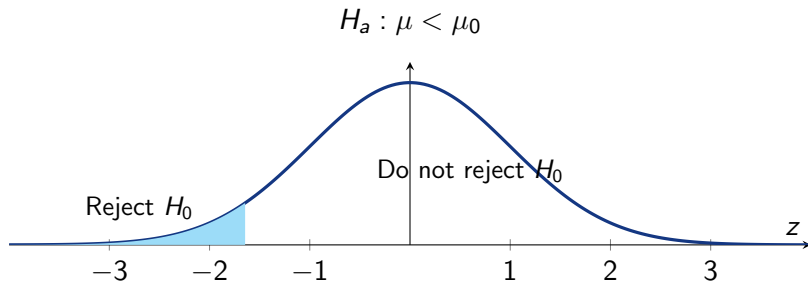
Tails of a Test

The tail depends on the alternative hypothesis.

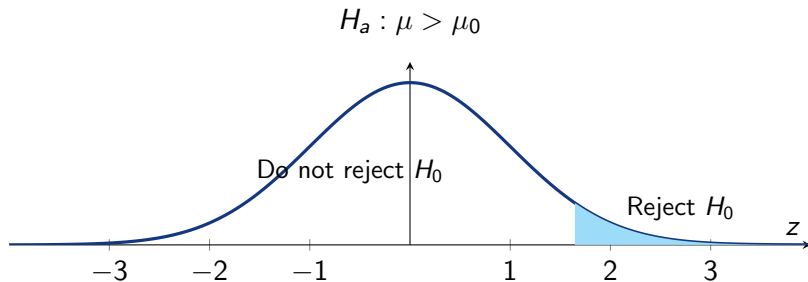
Alternative Hypothesis	Type of Test
$H_a : \mu < \mu_0$	Left-tailed
$H_a : \mu > \mu_0$	Right-tailed
$H_a : \mu \neq \mu_0$	Two-tailed

The sign in H_a tells you the tail.

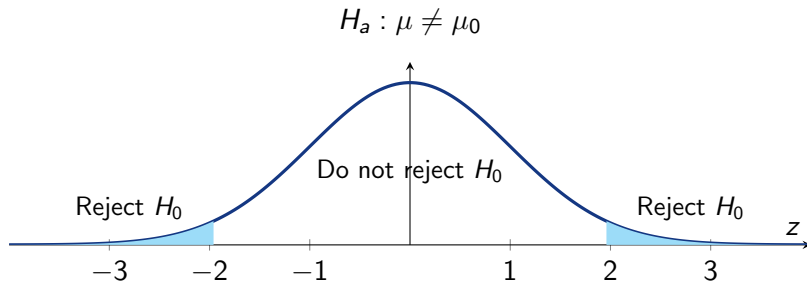
Left-Tailed Test



Right-Tailed Test



Two-Tailed Test



Quick Practice

Determine whether each test is left-tailed, right-tailed, or two-tailed.

- ① $H_0 : \mu = 80, \quad H_a : \mu < 80$
- ② $H_0 : \mu = 100, \quad H_a : \mu > 100$
- ③ $H_0 : \mu = 50, \quad H_a : \mu \neq 50$

- ① $H_a : \mu < 80$ left-tailed
- ② $H_a : \mu > 100$ right-tailed
- ③ $H_a : \mu \neq 50$ two-tailed

Again: always look at the sign in H_a .

Another Way to Make Decisions

So far, we have discussed:

- ▶ Null hypotheses
- ▶ Alternative hypotheses
- ▶ Rejection regions
- ▶ Critical values

Another common way to make decisions is by using the **p-value**.

The p-value

The **p-value** measures how unusual the sample result is if the null hypothesis is true.

Definition:

The p-value is the probability of obtaining a sample result at least as extreme as the observed result, assuming H_0 is true.

Interpretation:

- ▶ Small p-value $\Rightarrow$ strong evidence against H_0
- ▶ Large p-value $\Rightarrow$ little evidence against H_0

How Do We Use the p-value?

Compare the p-value to the significance level α .

Decision Rule:

$$\text{If } p \leq \alpha, \text{ reject } H_0$$

$$\text{If } p > \alpha, \text{ fail to reject } H_0$$

Common significance levels:

$$\alpha = 0.10, \quad 0.05, \quad 0.01$$

Example 1

Suppose

$$\alpha = 0.05$$

and the calculator reports

$$p = 0.012$$

Since

$$0.012 < 0.05,$$

we reject H_0 .

Conclusion:

There is sufficient evidence to support the alternative hypothesis.

Example 2

Suppose

$$\alpha = 0.05$$

and the calculator reports

$$p = 0.214$$

Since

$$0.214 > 0.05,$$

we fail to reject H_0 .

Conclusion:

There is not sufficient evidence to support the alternative hypothesis.

Important Notes

- ▶ A small p-value provides evidence against H_0 .
- ▶ A large p-value means the data are consistent with H_0 .
- ▶ We never say “accept H_0 .”
- ▶ We say:
 - Reject H_0
 - Fail to reject H_0

For any hypothesis test:

- ① State H_0 and H_a .
- ② Find the test statistic.
- ③ Find the p-value.
- ④ Compare the p-value to α .
- ⑤ Reject or fail to reject H_0 .
- ⑥ Write the conclusion in words.

Critical Values vs. p-values

Two common approaches to hypothesis testing:

Critical Value Approach

- ▶ Find critical value(s).
- ▶ Compare the test statistic to the critical value(s).
- ▶ Reject or fail to reject H_0 .

p-value Approach

- ▶ Find the p-value.
- ▶ Compare the p-value to α .
- ▶ Reject or fail to reject H_0 .

Both approaches lead to the same decision.

Section 9.2: Tests About μ When σ Is Known

In this section, we test claims about the population mean μ when the population standard deviation σ is known.

We use the **normal distribution** and the **critical value approach**.

Important

We are omitting the p-value approach.

When Do We Use a z-Test?

Use a z-test for the population mean when:

- ① The population standard deviation σ is known.
- ② The population is approximately normal, or the sample size is large:

$$n \geq 30.$$

The test statistic is:

$$z = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}$$

Meaning of the Symbols

$$z = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}$$

$\bar{x}$	sample mean
μ_0	hypothesized population mean from H_0
σ	population standard deviation
n	sample size
$\sigma / \sqrt{n}$	standard error

Critical Value Approach

A test of hypothesis using the critical value approach follows five steps.

Five Steps

- ① State the null and alternative hypotheses.
- ② Select the distribution to use.
- ③ Determine the rejection and nonrejection regions.
- ④ Calculate the value of the test statistic.
- ⑤ Make a decision.

Step 1: State the Hypotheses

Hypotheses are written about the population mean μ .

Examples:

$$H_0 : \mu = 50, \quad H_a : \mu \neq 50$$

$$H_0 : \mu \geq 300000, \quad H_a : \mu < 300000$$

$$H_0 : \mu \leq 22, \quad H_a : \mu > 22$$

The equality sign always belongs in H_0 .

Step 2: Select the Distribution

For Section 9.2, σ is known.

Therefore, we use the normal distribution.

The test statistic is:

$$z = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}$$

This is the same idea as standardizing, but now we standardize the sample mean.

Step 3: Determine the Rejection Region

The rejection region depends on the alternative hypothesis.

H_a	Test Type	Rejection Region
$\mu < \mu_0$	Left-tailed	$z < z_c$
$\mu > \mu_0$	Right-tailed	$z > z_c$
$\mu \neq \mu_0$	Two-tailed	$z < -z_c$ or $z > z_c$

Common Critical Values

α	Left-Tailed	Right-Tailed	Two-Tailed
0.10	-1.282	1.282	± 1.645
0.05	-1.645	1.645	± 1.960
0.01	-2.326	2.326	± 2.576

For a two-tailed test, split α equally between the two tails.

Step 4: Calculate the Test Statistic

Use:

$$z = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}$$

This tells us how many standard errors the sample mean is from the hypothesized mean.

Large positive or large negative values of z give stronger evidence against H_0 .

Step 5: Make a Decision

Compare the test statistic to the critical value.

- ▶ If the test statistic falls in the rejection region, reject H_0 .
- ▶ If the test statistic does not fall in the rejection region, fail to reject H_0 .

Then write a conclusion in the context of the problem.

Example 1: Long-Distance Calls

A telephone company claimed that the average length of long-distance calls was

12.44 minutes.

A sample of

$$n = 150$$

calls had

$$\bar{x} = 13.71$$

and the population standard deviation is known:

$$\sigma = 2.65.$$

At the 2% significance level, test whether the current mean length is different from 12.44 minutes.

Example 1: Step 1

The phrase **different from** tells us this is a two-tailed test.

$$H_0 : \mu = 12.44$$

$$H_a : \mu \neq 12.44$$

The significance level is:

$$\alpha = 0.02$$

Example 1: Step 2

Since σ is known, we use a z-test.

$$z = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}$$

Here:

$$\bar{x} = 13.71, \quad \mu_0 = 12.44, \quad \sigma = 2.65, \quad n = 150$$

Example 1: Step 3

This is a two-tailed test with

$$\alpha = 0.02.$$

Each tail has area

$$\frac{\alpha}{2} = \frac{0.02}{2} = 0.01.$$

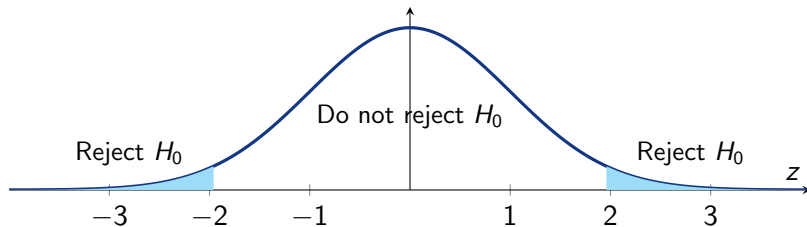
Critical values:

$$z = -2.33 \quad z = 2.33$$

Reject H_0 if

$$z \leq -2.33 \quad \text{or} \quad z \geq 2.33.$$

Example 1: Rejection Region



Reject H_0 if $z \leq -2.33$ or $z \geq 2.33$.

Example 1: Step 4

Compute the standard error:

$$\frac{\sigma}{\sqrt{n}} = \frac{2.65}{\sqrt{150}} \approx 0.2164$$

Then compute the test statistic:

$$z = \frac{13.71 - 12.44}{0.2164} \approx 5.87$$

Example 1: Step 5

The test statistic is

$$z = 5.87.$$

Since

$$5.87 > 2.33,$$

the test statistic falls in the rejection region.

Therefore, we reject H_0 .

Example 1: Conclusion

We reject H_0 .

There is sufficient evidence at the 2% significance level to conclude that the current mean length of long-distance calls is different from 12.44 minutes.

The sample mean is far enough from 12.44 that the difference is unlikely to be due to sampling error alone.

Example 2: Net Worth of Families

A mayor claims that the average net worth of families in a city is at least

$$\$300,000.$$

A random sample of

$$n = 25$$

families has

$$\bar{x} = \$288,000.$$

Assume the population standard deviation is

$$\sigma = \$80,000.$$

Use

$$\alpha = 0.025$$

Hypothesis Testing

Example 2: Step 1

The mayor claims the mean is at least \$300,000.

$$H_0 : \mu \geq 300000$$

$$H_a : \mu < 300000$$

This is a left-tailed test.

Example 2: Step 2

Since σ is known, we use a z-test.

$$z = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}$$

Here:

$$\bar{x} = 288000, \quad \mu_0 = 300000, \quad \sigma = 80000, \quad n = 25$$

Example 2: Step 3

This is a left-tailed test with

$$\alpha = 0.025.$$

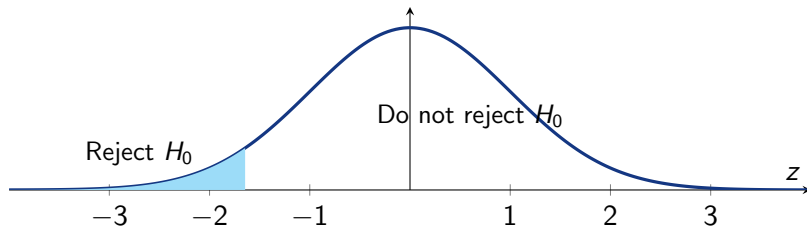
The critical value is:

$$z_c = -1.96.$$

Reject H_0 if

$$z \leq -1.96.$$

Example 2: Rejection Region



Reject H_0 if $z \leq -1.96$.

Example 2: Step 4

Compute the standard error:

$$\frac{\sigma}{\sqrt{n}} = \frac{80000}{\sqrt{25}} = 16000$$

Compute the test statistic:

$$z = \frac{288000 - 300000}{16000} = -0.75$$

Example 2: Step 5

The test statistic is

$$z = -0.75.$$

Since

$$-0.75 > -1.96,$$

the test statistic is not in the rejection region.

Therefore, we fail to reject H_0 .

Example 2: Conclusion

We fail to reject H_0 .

There is not sufficient evidence at the 2.5% significance level to conclude that the mean net worth of families in the city is less than \$300,000.

The sample evidence is not strong enough to reject the mayor's claim.

For a z-test, students may compute the test statistic directly:

$$z = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}$$

On the TI-84:

- ▶ Use parentheses carefully.
- ▶ Compute the denominator first if needed.
- ▶ Compare the final z value to the critical value.

For this lecture, we are not using the p-value decision method.

Summary: z-Test with Critical Values

Use a z-test when σ is known.

$$z = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}$$

Decision:

- ▶ Left-tailed: reject if z is too far left.
- ▶ Right-tailed: reject if z is too far right.
- ▶ Two-tailed: reject if z is too far in either direction.

Quick Check

A test has

$$H_0 : \mu = 40, \quad H_a : \mu > 40.$$

$$\alpha = 0.05.$$

- 1 Is this left-tailed, right-tailed, or two-tailed?
- 2 What is the critical value?
- 3 If the test statistic is $z = 1.91$, do we reject H_0 ?

Quick Check Answers

① Since $H_a : \mu > 40$, this is right-tailed.

② For $\alpha = 0.05$, the critical value is

$$z_c = 1.645.$$

③ Since

$$1.91 > 1.645,$$

we reject H_0 .

Section 9.3: When σ Is Unknown

In Section 9.2, the population standard deviation σ was known.

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$$

In Section 9.3, σ is unknown.

So we use the sample standard deviation s and the t-distribution.

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

When Do We Use a t-Test?

Use a t-test when:

- ▶ We are testing a claim about the population mean μ .
- ▶ The population standard deviation σ is unknown.
- ▶ The sample standard deviation s is given.

Degrees of freedom:

$$df = n - 1$$

Finding Critical t-Values on the TI-84

Use:

$2\text{nd} \rightarrow \text{VARs} \rightarrow \text{invT}$

The calculator needs:

- ▶ area to the left
- ▶ degrees of freedom

For this course, you may use the TI-84 instead of the t-table.

TI-84 Commands for Critical t-Values

Left-tailed test:

$$\text{invT}(\alpha, \text{df})$$

Right-tailed test:

$$\text{invT}(1-\alpha, \text{df})$$

Two-tailed test:

$$\text{invT}(1-\alpha/2, \text{df})$$

Then use positive and negative values:

$$\pm t_c$$

Five-Step Procedure

- 1 State H_0 and H_a .
- 2 Decide whether the test is left-tailed, right-tailed, or two-tailed.
- 3 Find the critical value using `invT`.
- 4 Compute the test statistic.
- 5 Reject or fail to reject H_0 .

Example: Children Starting Walking

A psychologist claims that the mean age at which children start walking is
12.5 months.

A sample of 18 children gives:

$$\bar{x} = 12.9, \quad s = 0.80$$

Use a 1% significance level to test whether the mean walking age is different from 12.5 months.

Example: Hypotheses

The phrase “different from” means this is a two-tailed test.

$$H_0 : \mu = 12.5$$

$$H_a : \mu \neq 12.5$$

$$\alpha = 0.01$$

Since σ is unknown, use a t-test.

$$df = n - 1 = 18 - 1 = 17$$

Example: Critical Values

This is a two-tailed test.

$$\alpha = 0.01$$

$$\frac{\alpha}{2} = 0.005$$

Use the TI-84:

$$\text{invT}(.995, 17)$$

Result:

$$t_c = 2.898$$

Critical values:

$$-2.898 \quad \text{and} \quad 2.898$$

Example: Test Statistic

Compute the standard error:

$$\frac{s}{\sqrt{n}} = \frac{0.80}{\sqrt{18}} \approx 0.1886$$

Compute the test statistic:

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$
$$t = \frac{12.9 - 12.5}{0.1886} \approx 2.121$$

Example: Decision

Critical values:

$$-2.898 \quad \text{and} \quad 2.898$$

Test statistic:

$$t = 2.121$$

Since

$$-2.898 < 2.121 < 2.898,$$

the test statistic is not in the rejection region.

We fail to reject H_0 .

Example: Conclusion

There is not sufficient evidence at the 1% significance level to conclude that the mean age at which children start walking is different from 12.5 months.

The sample mean is not far enough from 12.5 months to reject the claim.

Example: Bank Service

A bank previously served an average of 22 customers per hour.

After installing a new computer system, a sample of 70 hours gives:

$$\bar{x} = 27, \quad s = 2.5$$

Use a 1% significance level to test whether the new system is more efficient.

Bank Example: Hypotheses

More efficient means serving more customers per hour.

$$H_0 : \mu = 22$$

$$H_a : \mu > 22$$

This is a right-tailed test.

$$\alpha = 0.01$$

Since σ is unknown, use a t-test.

$$df = 70 - 1 = 69$$

Bank Example: Critical Value

Right-tailed test:

$$\alpha = 0.01$$

Use the TI-84:

$$\text{invT}(.99, 69)$$

Result:

$$t_c \approx 2.382$$

Reject H_0 if

$$t \geq 2.382.$$

Bank Example: Test Statistic

Compute the standard error:

$$\frac{s}{\sqrt{n}} = \frac{2.5}{\sqrt{70}} \approx 0.2988$$

Compute the test statistic:

$$t = \frac{27 - 22}{0.2988} \approx 16.733$$

Bank Example: Decision

Critical value:

$$t_c = 2.382$$

Test statistic:

$$t = 16.733$$

Since

$$16.733 > 2.382,$$

we reject H_0 .

There is sufficient evidence that the new computer system is more efficient.

Practice with invT

Find the critical value.

Left-tailed test:

$$\alpha = 0.05, \quad df = 15$$

TI-84:

$$\text{invT}(.05, 15)$$

Result:

$$t_c = -1.753$$

Practice with invT

Find the critical value.

Right-tailed test:

$$\alpha = 0.05, \quad df = 15$$

TI-84:

$$\text{invT}(.95, 15)$$

Result:

$$t_c = 1.753$$

Practice with invT

Find the critical values.

Two-tailed test:

$$\alpha = 0.05, \quad df = 15$$

TI-84:

$$\text{invT}(.975, 15)$$

Result:

$$t_c = 2.131$$

Critical values:

$$-2.131 \quad \text{and} \quad 2.131$$

Common Mistakes

- ▶ Writing hypotheses with $\bar{x}$ instead of μ .
- ▶ Putting the equality sign in H_a .
- ▶ Using z when σ is unknown.
- ▶ Forgetting that $df = n - 1$.
- ▶ Saying “accept H_0 ” instead of “fail to reject H_0 .”

z-Test or t-Test?

Use a z-test when:

σ is known.

Use a t-test when:

σ is unknown and s is given.

For Section 9.3, we use the t-test.

Chapter 9 Summary

For σ known:

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$$

For σ unknown:

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

Use the critical value approach:

Reject or fail to reject H_0 .

Final Roadmap

- 1 Write H_0 and H_a .
- 2 Decide the tail.
- 3 Decide z-test or t-test.
- 4 Find the critical value.
- 5 Compute the test statistic.
- 6 Make the decision.
- 7 Write the conclusion in context.